

PROJECT PRESENTATION

Digital Certificate Verification System

A Java Swing desktop application for issuing, managing and verifying digital certificates

CERTIFICATES

Issue & store

VERIFY

Authenticity check

SEARCH

ID, name, recipient

REPORTS

Sort, history, stats

Introduction & Problem Statement

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Why the system is needed

PROBLEM STATEMENT

- Paper certificates are easy to forge.
- Records are hard to search and track.
- Verification by hand is slow and error-prone.
- Duplicate certificates must be prevented.

PROPOSED SOLUTION

- A simple Java Swing desktop application.
- Unique certificate ID generated per recipient and course.
- HashMap gives instant lookup by ID.
- Validation blocks bad input and duplicates.

CORE OPERATIONS

Generate

Verify

Search

Update / Delete

GUI Components

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Implementation technologies and interface elements

SWING COMPONENTS

- JFrame: main application window
- JPanel: groups form and buttons
- JTextField: accepts certificate data
- JButton: triggers the 11 operations
- JTextArea: displays system output
- JScrollPane: scrollable output
- JOptionPane: errors, input, confirm

The screenshot shows a Java Swing window titled "Digital Certificate Verification System". The window has a title bar with standard macOS window controls. The main content area is divided into two sections. The top section, titled "Certificate Details", contains a form with five input fields: "Recipient ID", "Recipient Name", "Email", "Course Code", and "Certificate ID". Below these fields is a "Search" button. The bottom section, titled "System Output", contains a large text area for displaying system output. Above the text area, there is a row of buttons: "Generate", "Verify", "Search", "Update", "Delete", "Show All", "Sort by Date", and "Sort by Co...".

Main window of the application

Application Interface

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Main window of the Java Swing application

INTERFACE ELEMENTS

- Title label and a titled "Certificate Details" form (GridLayout 4x4).
- Fields: Recipient ID, Name, Email, Course Code, Course Name, Search.
- Certificate ID is read-only and auto-filled.
- Button panel at the top, output area below.

USER INTERACTION

- Generate: create a certificate
- Verify: check authenticity
- Search: find by ID, recipient ID or name
- Update / Delete: modify or remove
- Show All, Sort by Date, Sort by Course
- History, Report and Clear

Class Design (OOP)

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Classes and the interface behind the system

Verifiable (interface)

Declares verify() returning boolean.

Recipient

Holds id, name and email.

Course

Holds course code and name.

Certificate

Implements Verifiable. Has id, Recipient, Course and issue date.

Verifier

check() returns VERIFIED or INVALID message.

Main class (JFrame)

Builds the GUI, holds collections and all operations.

Data Structures

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How each collection supports a specific requirement

ARRAYLIST: ALL CERTIFICATES

- Master list of Certificate objects.
- Sorted with Comparator.

HASHMAP: ID SEARCH

- Certificate ID = key.
- `get(id)` returns the certificate.

LINKEDLIST: HISTORY

- Stores issuance records.
- Items added in order of issue.

TREEMAP: DATE REPORT

- `LocalDate` = key, sorted.
- Value = list of certificates.

Certificate Generation: Program Flow

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Validation occurs before a certificate is stored



CERTIFICATE ID FORMAT

CERT-<COURSE CODE>-<3-digit number> e.g. CERT-JAVA101-001

Verify, Update & Delete

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How existing certificates are checked and changed

VERIFY

- Takes ID from the field or Search box.
- HashMap lookup by ID.
- Verifier.check() calls verify().
- Shows VERIFIED or INVALID.

UPDATE

- Needs a selected certificate ID.
- Input dialogs for new name and email.
- Email checked for @.
- Recipient object is updated.

DELETE

- Confirmation dialog first.
- Removed from ArrayList and HashMap.
- Removed from history and date map.
- Empty date entries are cleaned up.

Display, Sorting & Reports

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Show All, Sort, History, Report and Clear

SHOW ALL & SEARCH

- Show All lists every certificate.
- Search matches ID, recipient ID or name.

SORTING

- Sort by Date uses Comparator on issue date.
- Sort by Course compares course name.

HISTORY & REPORT

- History shows the issuance log.
- Report: total, valid, invalid, count per date.

CLEAR

- Resets all fields and the output area so the form is ready for the next operation.

Conclusion

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A complete Java Swing solution for certificate issue and verification

KEY OUTCOMES

- ArrayList manages all certificates.
- HashMap enables ID-based lookup.
- LinkedList keeps issuance history.
- TreeMap groups certificates by date.
- OOP: interface, classes, encapsulation.

SYSTEM WORKFLOW

- User enters details and clicks Generate.
- Input is validated and checked for duplicates.
- Certificate is stored with a unique ID.
- Verify, search, update or delete as needed.
- Report summarises all certificates.

THANK YOU • QUESTIONS ARE WELCOME